

Cross-Reader Generality of PAQ-KV: A Multi-VLM Study (Internal Companion Note)

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Abstract

PAQ-KV is a training-free, index-free, single-reader method: a vision-language model (VLM) prefills a multi-page document once, and per query computes its *own* question-to-context attention, selects the top $\approx 5\%$ of 64-token visual key/value (KV) blocks, and decodes under a reversible 4-D restricted-attention mask over its own KV cache. Its headline result—beating a trained ColQwen2 retriever on MMLongBench-Doc accuracy—is established on Qwen3-VL-8B and confirmed out-of-sample. This note asks the natural follow-up: *does the advantage generalize to other VLM readers?* We port the full method (reader-own scoring, selection, and masked decode) to additional VLMs and report matched-budget results on a fixed 80-document / 444-query development (DEV) pin. Findings so far: on Qwen2.5-VL-7B (same Qwen-VL family) the method is a *statistical dead heat* with the comparator ($\Delta = +0.007$, doc-cluster $LB_{95} = [-0.037, +0.049]$); on InternVL3.5-8B (a different VLM whose language backbone is Qwen3) it *significantly underperforms* ($LB_{95} = [-0.106, -0.029]$). Together these characterize a **mechanistic boundary**: the method’s benefit tracks how good the host reader’s own attention is as a below-page retriever. We are extending the study to four further readers (Qwen3-VL-4B, Molmo-7B-D, MiniCPM-V-4.5, PaliGemma-2). This is an internal companion note kept separate from the submission; the shared team decides what, if anything, to fold in.

1 Setup

1.1 The method and the established result

PAQ-KV prefills a full multi-page document once into a VLM’s KV cache. For each query it (i) reads the reader’s own question-to-context attention (scorer v0: mean over heads and layers), (ii) partitions the retained visual KV into 64-token stripe blocks and selects the top $\approx 5\%$ by attention score, and (iii) decodes the answer under a reversible 4-D additive attention mask that hides all non-selected visual columns while always keeping the (small) non-visual scaffold. It trains nothing, builds no external index, and uses a single model.

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On **Qwen3-VL-8B**, out-of-sample on a sealed 45-document / 244-query MMLongBench-Doc split, PAQ-KV at 5% scores **0.3957** versus a trained ColQwen2 retriever’s **0.3202** ($+0.0755$; doc-cluster $LB_{95} = +0.0290$). On the 80-document / 444-query DEV pin the same beat is $+0.079$ ($LB_{95} = +0.041$). This note does not touch that result; the sealed split remains evaluated exactly once.

1.2 What generality requires of a reader

Porting surfaced hard prerequisites that any candidate reader must satisfy for the method to even run:

1. **White-box access.** The method reads attention weights, edits the KV cache, and injects a 4-D mask. Any API / closed-weights model is categorically impossible.
2. **In-line visual tokens in the decoder’s self-attention KV.** Blocks are partitions of visual token *positions* in the *self-attention sequence*. A reader that injects vision via *cross-attention* (vision lives outside the self-attention KV) has no visual KV blocks to select or mask.
3. **Hookable eager attention returning 4-D weights**, plus a **4-D additive mask honored by the language model** (the transferability crux; native TRANSFORMERS classes that delegate to a native decoder satisfy this, custom `trust_remote_code` attentions may not).
4. **Reconstructable, continuable positions** (1-D arrange or M-RoPE) and enough context to prefill a multi-page document. Sub-native context ceilings force a reduced-budget setting (a disclosed confound, §4).
5. Empirically (§3), **the reader’s own attention must actually be a good below-page retriever**. Mechanical portability is necessary but not sufficient.

1.3 Protocol

All second-reader numbers are on the fixed **80-document / 444-query DEV pin** (`paq_confirm_pin.json`, MMLongBench-Doc DEV), reported as a *generality control*, not a new sealed claim. Each reader is compared to itself under two selectors at a matched budget: PAQ-KV (own attention selects $\approx 5\%$ of blocks) vs. a *comparator* arm in which the same reader reads the top-5% pages chosen by cached ColQwen2 retrieval scores (no new ColQwen2 runs). We also record a *full-context* reference. The primary statistic is the doc-cluster 95% lower bound

LB₉₅ of (paqkv – comparator) (seed 0, 10,000 bootstrap resamples); the pre-registered “transfers” bar is LB₉₅ > 0. An adversarial worst-case imputation of any missing query binds the verdict. A hard **identity gate** (teacher-forced argmax parity of the all-true 4-D mask vs. native causal decode, plus a restrictive mask changing the output, on ≥2 distinct documents including one ≥30 pages) must pass before any number is trusted. Verdicts are labeled INCOMPATIBLE (identity fails / port infeasible), INCOMPLETE (coverage/power short), READER2_NO_TRANSFER (LB₉₅ ≤ 0 with clean coverage), or READER2_TRANSFERS (LB₉₅ > 0).

2 Results

2.1 Main comparison (verified readers)

Table 1 reports the three verified readers on the same 444-query DEV pin. Accuracy is the MMLongBench-Doc scorer; higher is better. All three readers read the same content selectors (own PAQ-KV blocks vs. ColQwen2 pages) at a matched ≈5% budget.

- **Qwen3-VL-8B (native reader): win.** PAQ-KV – ColQwen2 = +0.076, LB₉₅ = +0.041; PAQ-KV even edges its own full context (0.4042 vs. 0.3965), i.e. selection *denoises*.
- **Qwen2.5-VL-7B (same Qwen-VL family): tie.** Δ = +0.0074, LB₉₅ = [−0.037, +0.049]—a statistical dead heat straddling zero (PAQ-KV nominally ahead). Verdict READER2_NO_TRANSFER by the LB₉₅ > 0 bar, but this is “no detectable difference,” not a loss. Per-query: 54 wins / 331 ties / 59 losses. On the 128 queries answerable by full context, PAQ-KV recovers **0.703** vs. the comparator’s **0.581**—here PAQ-KV is the *better* below-page selector. It loses only −0.054 [−0.087, −0.024] to its own full context.
- **InternVL3.5-8B (cross-family, Qwen3 LM): significant negative.** PAQ-KV – ColQwen2 = −0.067, LB₉₅ = [−0.106, −0.029] (whole interval below zero). It also trails its own full context (−0.076 [−0.113, −0.041]). On answerable queries it recovers only **0.529**, losing ≈47% of them to its 8% selection. Verdict READER2_NO_TRANSFER (a genuine, well-powered negative).

2.2 Additional readers (in progress)

Table 2 records the four further readers requested for this study, with the verified architecture facts that determine feasibility (§1.2) and the current run status. Feasibility is decided cheaply by a Phase-0 identity spike before any full run; full ports proceed only for spike-passers, and a reader that fails the identity gate is reported honestly as INCOMPATIBLE rather than forced.

Per-reader notes.

- **Qwen3-VL-4B** is the same architecture as the sealed 8B (same TRANSFORMERS class, M-RoPE, native per-image budget), so the existing harness runs with a loader swap. It is a *within-architecture size ablation*, not a cross-architecture test; a win here would say the method sur-

vives shrinking the same reader, not that it transfers across families.

- **Molmo-7B-D** uses a Qwen2 language backbone with LLaVA-style in-line visual tokens (genuinely different training data, AI2 PixMo), but ships as trust_remote_code; whether its custom attention honors the 4-D mask and exposes 4-D eager weights is the spike’s make-or-break. If it ports, it is another Qwen2-family data point (cf. Qwen2.5-VL).
- **MiniCPM-V-4.5** has a Qwen3 backbone (the same LM family as InternVL, which was negative) and a 3D-Resampler that compresses each image to ≈64 tokens; at ≈64 visual tokens per page, a 5% block budget leaves almost nothing to select below page granularity, so even a clean port is expected to be near-degenerate. It is also trust_remote_code.
- **PaliGemma-2** uses a prefix-LM (bidirectional attention over the image+prefix, causal only on the suffix), which conflicts with the method’s causal restricted-decode over a prefilled cache; it is also short-output-tuned and weakly suited to long multi-page VQA. The spike will decide between INCOMPATIBLE and a genuine-but-weak result.

3 Discussion: a mechanistic generality gradient

The verified readers order cleanly by architectural distance from the reader the method was developed on (Table 3). The benefit is not universal; it *degrades monotonically* as the reader moves away from Qwen3-VL-8B, and the degradation tracks a measurable property—how well the reader’s own question-to-context attention localizes the answer below page granularity. Qwen3-VL-8B has this property (it recovers answerable queries well at 5% and even denoises above full context); Qwen2.5-VL-7B partially has it (break-even; recovers 0.70 of answerable queries); InternVL3.5-8B lacks it here (recovers 0.53 and loses). This *confirms* rather than refutes the single-reader observation, and reframes it from an unexamined weakness into an investigated boundary condition: PAQ-KV is a white-box method whose benefit is contingent on the host reader’s attention being an effective below-page retriever.

4 Confounds and honesty caveats

Any use of these numbers must carry the following.

1. **Reduced-budget confound.** Both second readers run below the native-long 8B budget (InternVL ~36K via a 40K context ceiling; Qwen2.5-VL ~52K via a 128K ceiling plus a processor pre-resize). Within-reader matched-budget comparisons carry the findings; cross-reader comparisons *against* the 8B are confounded by capacity and budget and are secondary only.
2. **Shared LM family.** InternVL3.5-8B’s language backbone is Qwen3, which is *why* the KV/attention/mask machinery transfers. It is a legitimate cross-VLM result but not a fully independent language architecture; disclose it.
3. **Conservative-against-method budget asymmetry (Qwen2.5-VL).** The method downscales all pages in the

Table 1: PAQ-KV vs. the ColQwen2-page comparator, within each reader, on the 80-doc / 444-q DEV pin. Δ and LB_{95} are doc-cluster (paqkv – comparator). Qwen3-VL-8B is the reader the method was developed and sealed on.

Reader	Budget	PAQ-KV	ColQwen2	Full	doc-cluster LB_{95} (paqkv – ColQwen2)
Qwen3-VL-8B (<i>native</i>)	~64K	0.4042	0.3284	0.3965	+0.041
Qwen2.5-VL-7B	~52K	0.2698	0.2623	0.3237	[−0.037, +0.049]
InternVL3.5-8B	~36K	0.1876	0.2542	0.2637	[−0.106, −0.029]

Table 2: Additional candidate readers. “Native” = a TRANSFORMERS 5.5 class exists (mask machinery and eager hooks transfer as for InternVL/Qwen2.5-VL); “remote” = trust_remote_code custom attention (4-D mask honored and hookability must be validated by the spike). Results are filled in as runs complete.

Reader	LM backbone	Vision integration	Impl.	Expected	Status
Qwen3-VL-4B	Qwen3-VL	M-RoPE, in-line tokens	native	win-ish (size abl.)	<i>pending</i>
Molmo-7B-D	Qwen2-7B	CLIP, in-line tokens	remote	tie-ish (Qwen2 fam.)	<i>pending</i>
MiniCPM-V-4.5	Qwen3-8B	3D-Resampler (64 tok)	remote	degenerate/neg.	<i>pending</i>
PaliGemma-2	Gemma-2	prefix-LM (bidir.)	native	incompat./weak	<i>pending</i>

full-document prefill, whereas the comparator renders its 1–5 kept pages near-native and thus gets a slightly larger per-page budget. The tie is therefore a floor on the within-family effect.

4. **DEV-pin, not a new sealed claim.** All second-reader numbers are on the DEV pin as a generality control. The sealed split is untouched.

Verification (Appendix A)

Every reported run passed a hard identity gate and was independently checked. For each verified reader: the identity audit confirmed the all-true 4-D mask reproduces native causal decoding under the teacher-forced argmax criterion ($\max |\Delta \logit| = 0$), a restrictive mask changes the output, on ≥ 2 distinct documents (one ≥ 30 pages), with the resolved model revision and code hash bound into the frozen run manifest. A CPU sparse-mask audit confirmed the block-to-column mapping is exact (all-blocks selection reproduces the all-true mask; a sparse subset masks exactly the intended columns). An off-meter code reviewer independently attempted to falsify each result and found no implementation defect (“genuine negative” for InternVL; “genuine null with favorable caveats” for Qwen2.5-VL). Full coverage for both second readers: 444/444 queries, 80 documents, 0 skipped, 0 dropped, identity pass. Artifacts, verdict JSONs, reviewer transcripts, and audit scripts are in the project repository under research/ and research/reviews/paq_second_vlm_2026-07-22/; the consolidated running log is research/design_notes/paq_cross_reader_generality_2026-07-23.md.

References (model / benchmark sources)

Qwen3-VL / Qwen2.5-VL Qwen Team. <https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct>

InternVL3.5-8B OpenGVLab. https://huggingface.co/OpenGVLab/InternVL3_5-8B-HF

Molmo-7B-D Allen Institute for AI. <https://huggingface.co/allenai/Molmo-7B-D-0924>

MiniCPM-V-4.5 OpenBMB. https://huggingface.co/openbmb/MiniCPM-V-4_5

PaliGemma-2 Google. <https://huggingface.co/google/paligemma2-3b-mix-448>

ColQwen2 Faysse et al. <https://huggingface.co/vidore/colqwen2-v1.0>

MMLongBench-Doc Ma et al. <https://mmlongbench-doc.github.io>

Table 3: The generality gradient. doc-cluster $LB_{95}(\text{paqkv} - \text{comparator})$; Qwen3-VL-8B shows the sealed out-of-sample value, the second readers the DEV-pin value.

Reader	Relation to the sealed reader	doc-cluster LB_{95}
Qwen3-VL-8B	native (sealed reader)	+0.029 (<i>win</i>)
Qwen2.5-VL-7B	same family, prior gen.	$[-0.037, +0.049]$ (<i>tie</i>)
InternVL3.5-8B	cross-family VLM (Qwen3 LM)	$[-0.106, -0.029]$ (<i>neg.</i>)